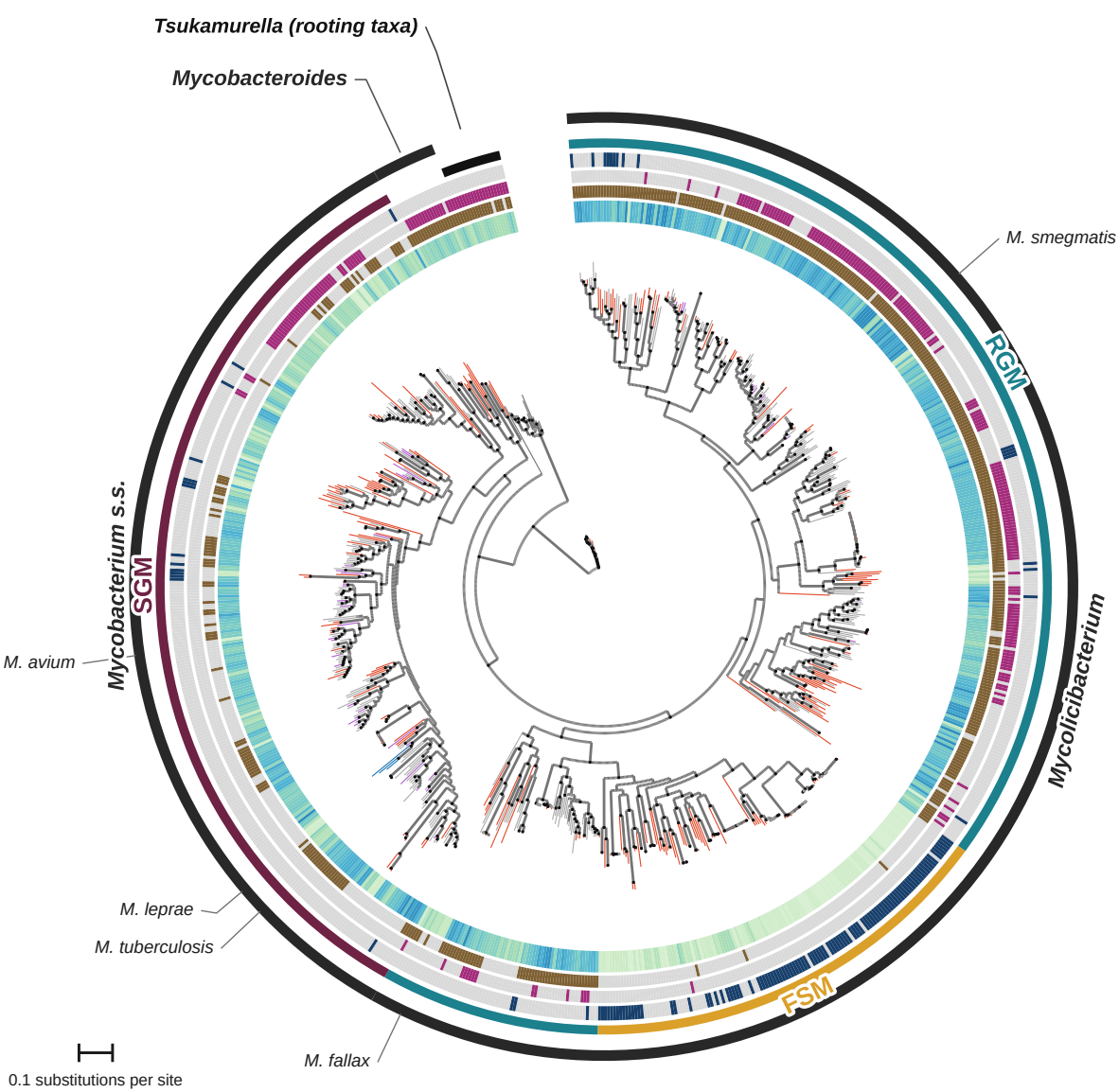


Clade (arcs)

- SGM slow-growing clade
- RGM large-genome grade (paraphyletic)
- FSM small genome, mainly freshwater

Node support (UFBoot)

- ● ≥ 95
- ● 90–94
- < 90: no dot

Branch colour (tip category)

- Cultured isolate
- NTM isolate (cultured)
- MAG: 246 SPIRE + 182 GTDB refs
- Tissue-derived (never cultured)
- Rooting taxa (Tsukamurella)
- Internal branches; 19 basal tips*

* Mycobacteroides-grade genomes outside the 824-genome crown; always grey.

Ring 1 Genome size (Mb)

≤ 2.9 — ≥ 7.4

Rings 2–4 Gene presence

- Urease
- Ectoine synthase
- Microbial rhodopsin (opsin gene)
- Absent

862 genomes: 246 SPIRE MAGs + 616 GTDB references (182 of them MAGs, in the MAG colour). Radial scale: substitutions per site above the crown of the 843 non-Tsukamurella genomes (stem compressed). Clade arcs cover the 824-genome crown only. GC: Fig. 2a; proteome acidity: Tables S14, S1.